

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q15-0538

Gordonsville Energy Lp 138 kV
December 17, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0538. The Interconnection Customer proposes a 157.8 MW solar photovoltaic facility interconnecting at the Gordonsville Energy Lp 138 kV substation in the Virginia area. The request was received on November 13, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 183 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$11,780,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
LITTLEFORK-HOGAN RD 138.0 kV Line	LOUISA GENERATION FACILITY-NORTH ANNA 345.0 kV	142.4	271	9.5
MUSTANG-ELECTRIC AVENUE FACILITY 138.0 kV	HOGAN RD-TUCKASEGEE TIE SUBSTATION 138.0 kV	119.3	246	5.5
STATE T-ROBISON PARK 230.0 kV	DEKALB-RICHLAND 500.0 kV	118.0	1,378	6.1
MOUNT PLEASANT-FIREBANTZ 138.0 kV	LOUISA GENERATION FACILITY 345/138 kV Autotransformer	112.9	236	5.7
LITTLEFORK-COVANTA SOUTHEASTERN CONNECTICUT COMPANY 138.0 kV	PRATT'S-MUSTANG 138.0 kV Line	110.6	246	6.9
SPRING GAP TAP-COVANTA SOUTHEASTERN CONNECTICUT COMPANY 138.0 kV	MUSTANG-PRATT'S 138.0 kV	110.4	246	6.9
ROBISON PARK SUBSTATION-RUSSELL FORK 230.0 kV	DEKALB-JONES CHAPEL 500.0 kV	107.1	1,695	7.8
DEKALB SUBSTATION 500-1 kV Autotransformer	RICHLAND-DEKALB 500.0 kV	104.7	1,488	6.6
WHEELER-FIREBANTZ 138.0 kV Line	SOUTH JORDAN-CHAFFEE GAS RECOVERY SUBSTATION 138.0 kV	104.4	349	5.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LITTLEFORK-HOGAN RD 138.0 kV	Rebuild line with higher-capacity conductor	\$9,907,000	23% (\$2,304,000)
MUSTANG-ELECTRIC AVENUE FACILITY 138.0 kV	Rebuild line with higher-capacity conductor	\$12,017,000	8% (\$898,000)
STATE T-ROBISON PARK 230.0 kV	Rebuild line with higher-capacity conductor	\$9,274,000	1% (\$109,000)
MOUNT PLEASANT-FIREBANTZ 138.0 kV	Rebuild line with higher-capacity conductor	\$5,924,000	11% (\$654,000)
LITTLEFORK-COVANTA SOUTHEASTERN CONNECTICUT COMPANY 138.0 kV	Rebuild line with higher-capacity conductor	\$23,120,000	25% (\$5,869,000)
SPRING GAP TAP-COVANTA SOUTHEASTERN CONNECTICUT COMPANY 138.0 kV	Rebuild line with higher-capacity conductor	\$4,759,000	25% (\$1,208,000)
ROBISON PARK SUBSTATION-RUSSELL FORK 230.0 kV	Reconductor line	\$23,069,000	1% (\$246,000)
DEKALB SUBSTATION 500-1 kV Autotransformer	Replace transformer with higher-rated unit	\$23,978,000	1% (\$256,000)
WHEELR-FIREBANTZ 138.0 kV	Reconductor line	\$2,293,000	10% (\$236,000)
	Total allocated to project		\$11,780,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.